

Obesity in Hidalgo County

A Data-Driven Public Health Analysis

Hidalgo County, Texas

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Abstract

Hidalgo County, Texas ranks in the 94th percentile nationally for obesity, with a crude prevalence rate of 44.2% — nearly 7 percentage points above the national average of 37.4%. This report analyzes county-level public health data from the CDC PLACES dataset (2024) to identify the structural drivers of this elevated burden, assess their statistical significance, and model the potential impact of targeted public health interventions. Using a combination of statistical testing, machine learning, and Monte Carlo simulation, we find that physical inactivity, sleep deprivation, unmanaged diabetes, and lack of health insurance are the four variables where Hidalgo most significantly exceeds the national average — and that simultaneously reducing all four to national levels could reduce obesity prevalence by an estimated 7.18 percentage points. These findings have direct implications for clinical practice and public health policy in Hidalgo County.

1. Introduction

The Rio Grande Valley (RGV) is a cluster of counties along the southernmost tip of Texas bordering Mexico. This region of Texas is often considered one of the most economically disadvantaged regions in the United States. Among its counties, Hidalgo County stands out not only for its size but also for its disproportionately high burden of chronic disease, particularly obesity and its downstream consequences.

Obesity is not simply a matter of individual lifestyle choices. It is a complex, multifactorial condition shaped by access to healthy food, opportunities for physical activity, quality of sleep, access to healthcare, socioeconomic stress, and genetic predisposition. In communities like Hidalgo County, these risk factors do not occur in isolation, but rather cluster together, creating a compounding burden that far exceeds what any single intervention can address.

This report asks three core questions:

- Is the claim that Hidalgo County is among the most obese regions in the United States statistically substantiated?
- Which specific risk factors are most strongly driving elevated obesity rates in Hidalgo County relative to the rest of the country?
- If providers and public health officials successfully intervened on those risk factors, what reduction in obesity rates could realistically be expected?

The answers to these questions have real consequences. Providers in the RGV see the effects of chronic obesity daily. Public health officials must allocate limited resources across competing priorities. This report is designed to give both groups the data they need to make evidence-based decisions.

One might ask, why does this matter right now? Obesity rates in the United States have continued to rise over the past two decades, but the burden has not been evenly distributed. Counties like Hidalgo have seen disproportionate increases in obesity. Without targeted intervention, the gap between high-burden and low-burden counties will continue to widen, with direct consequences for healthcare costs, workforce productivity, and quality of life.

2. Data Source: CDC PLACES Dataset

All data used in this analysis come from the CDC PLACES (Population Level Analysis and Community Estimates) dataset, 2024 release. PLACES provides model-based estimates of health measures at the county, city, census tract, and ZIP code levels for every county in the United States; as a result, there are 3,143 rows in this data set. This is important because traditional health surveys only collect data at the state level, leaving local health departments and providers without county-specific information. PLACES fills that gap.

PLACES uses a statistical technique called Multilevel Regression and Poststratification (MRP). In plain words, the CDC collects health survey responses from hundreds of thousands of Americans through the Behavioral Risk Factor Surveillance System (BRFSS), then uses sophisticated modeling to estimate what the rates would be in every county, accounting for the demographic composition of each county using U.S. Census data. The result is a reliable estimate for each health measure in each county.

CDC PLACES is published by a federal agency, peer-reviewed, and updated annually. It is widely used in academic research, clinical planning, and public health policy. Its county-level estimates have been validated against direct survey estimates and show strong correlations at the state and county levels. For county-level comparative analysis, it is the most comprehensive and reliable publicly available source.

The 2024 PLACES release contains 40 health measures across six categories for 3,143 U.S. counties. For this analysis, we selected 8 variables most relevant to obesity and its structural drivers. The dataset was accessed directly via the CDC's public API and required no special authorization.

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3. Variables of Interest

The following variables were selected from the PLACES dataset. Each is expressed as a crude prevalence rate, which is the percentage of adults in that county estimated to have that characteristic or condition. All values represent adults aged 18 and older.

Variable	Plain English	Why it matters for obesity
obesity_crudeprev	% of adults with obesity (BMI ≥ 30)	Primary outcome variable

lpa_crudeprev	% of adults with no leisure-time physical activity	Sedentary behavior is one of the strongest modifiable predictors of obesity
diabetes_crudeprev	% of adults diagnosed with type 2 diabetes	Obesity and diabetes are tightly linked
access2_crudeprev	% of adults with no health insurance	Uninsured adults receive less preventive care and chronic disease management
bphigh_crudeprev	% of adults with high blood pressure	Cardiovascular comorbidity is closely associated with obesity
csmoking_crudeprev	% of adults who currently smoke	Shares behavioral risk factors with obesity
sleep_crudeprev	% of adults sleeping less than 7 hours/night	Short sleep disrupts hunger hormones and is associated with weight gain
depression_crudeprev	% of adults with diagnosed depression	Depression reduces motivation for activity, possibly affecting obesity
cholscreen_crudeprev	% of adults who received cholesterol screening	Proxy for healthcare access; higher screening = better preventive care engagement

4. Methodology

This analysis combines classical statistics with modern machine learning. Each method answers a different question, and together they build a comprehensive picture of obesity in Hidalgo County. A justification for each method is provided below.

4.1 Descriptive Analysis

What it does: Calculates the average value of each health variable for Hidalgo County, all Texas counties, and all U.S. counties, then places them side by side for comparison.

Why it is valid: Descriptive statistics are the foundation of any public health analysis. Before asking why something is happening, one must first establish what is happening and how it compares to relevant benchmarks.

4.2 One-Sample T-Test

What it does: Tests whether Hidalgo County's obesity rate is statistically significantly different from the distribution of obesity rates across all Texas counties or all U.S. counties.

What 'statistically significant' means: In science, we never say something is 100% true, but we can say it is unlikely to be due to random chance. A p-value is the probability that the difference we observed could have occurred by random chance alone. A p-value below 0.05 is the conventional threshold for calling a result statistically significant.

Why it is valid: The one-sample t-test is the appropriate test when comparing a single observation (Hidalgo County) against a population distribution (all U.S. counties).

4.3 Linear Regression

What it does: Builds a mathematical formula that predicts a county's obesity rate based on its values for the 8 health variables. The model is trained on 80% of counties and tested on the remaining 20%.

Why it is valid: The R^2 statistic produced by the model tells us what proportion of the variation in obesity rates across counties is actually explained by the model.

4.4 Logistic Regression, Decision Tree & Random Forest

What they do: These models classify counties as either 'high obesity' (above the national median) or 'low obesity' based on their health variable profiles. The national median obesity rate was 37.9%.

Why they are valid: Classification models are used when the goal is to identify which group an observation belongs to. The ROC curve and AUC metric measure how well the model distinguishes between high and low obesity counties. An AUC of 0.88 (as achieved here) is considered strong performance. Random Forest, which combines hundreds of decision trees, is particularly robust because it reduces overfitting and provides reliable feature importance scores.

4.5 Principal Component Analysis (PCA) & K-Means Clustering

What they do: PCA compresses the 8 health variables into 2 summary dimensions that capture the most important variation. K-Means then groups all 3,143 counties into 4 clusters based on their overall health profiles. This reveals which type of county Hidalgo most resembles.

Why they are valid: When variables are correlated (smoking, inactivity, and high blood pressure tend to occur together), PCA removes that redundancy before clustering, producing more meaningful groups. These methods are standard in public health surveillance and health geography research.

4.6 Monte Carlo Simulation

What it does: Runs 10,000 simulations in which one or more of Hidalgo's elevated risk factors is replaced with a random draw from the national distribution, while all other factors are held constant. The result is a range of predicted obesity rates — showing what the model would expect if that risk factor were reduced to typical national levels.

Why it is valid: These models essentially show a 'what if' scenario. By running 10,000 simulations, we obtain a stable probability distribution of outcomes rather than a single point estimate. The 95% confidence interval produced tells us the range within which we would expect the true result to fall 95% of the time. This is a more honest and rigorous approach than simply plugging in the national average once.

5. Descriptive Analysis

The table below presents Hidalgo County's crude prevalence rates for each health variable, along with Texas and national averages.

Variable	Hidalgo Co.	Texas Avg	National Avg	vs. National
Obesity (%)	44.2	37.4	37.4	+6.8%

Diabetes (%)	17.4	14.6	13.6	+3.8%
Physical Inactivity (%)	37.5	31.3	28.5	+9.0%
No Insurance (%)	37.4	20.4	11.1	+26.3%
High Blood Pressure (%)	36.2	37.7	38.7	-2.5%
Smoking (%)	11.5	14.7	15.8	-4.3%
Short Sleep (%)	40.4	38.2	35.8	+4.6%
Depression (%)	14.8	21.0	22.1	-7.3%
Cholesterol Screening (%)	73.1	77.2	79.4	-6.3%

Findings: Hidalgo County exceeds the national average on four variables by substantial margins: physical inactivity (+9.0 percentage points), no health insurance (+26.3 pp), short sleep (+4.6 pp), and diabetes (+3.8 pp). Notably, Hidalgo has lower smoking and depression rates than the national average — suggesting that the obesity burden here is driven primarily by structural and metabolic factors rather than behavioral ones like tobacco use.

6. Statistical Testing

To determine whether Hidalgo County's obesity rate is significantly different from Texas and national averages, we conducted one-sample t-tests for both Hidalgo and Cameron County.

Comparison	t-statistic	p-value	Significant?
Hidalgo vs. Texas counties	-37.246	< 0.0001	Yes ✓
Hidalgo vs. All U.S. counties	-77.495	< 0.0001	Yes ✓
Cameron vs. Texas counties	-15.811	< 0.0001	Yes ✓
Cameron vs. All U.S. counties	-32.894	< 0.0001	Yes ✓

All four comparisons are statistically significant at $p < 0.0001$ — meaning there is less than a 1 in 10,000 probability that these differences occurred by chance. The large negative t-statistics reflect both the magnitude of the difference and the large sample size (3,143 counties nationally). Hidalgo County's obesity rate is unambiguously elevated relative to all relevant benchmarks.

National ranking: Hidalgo County ranks 153rd out of 3,143 U.S. counties for obesity prevalence — placing it in the 94th percentile nationally. Cameron County ranks 782nd (73rd percentile). This confirms that Hidalgo County seems to be a more prominent driver of the region's elevated burden.

7. Linear regression model

The model was trained on 2,365 counties and tested on 591 counties it had never seen before, producing the following results:

- **R² Score: 0.6595:** This means the 8 health variables together explain approximately 66% of the variation in obesity rates across all U.S. counties. An R² of 0.66 at the county level is considered strong. The remaining 34% of variation is likely attributable to factors not captured in this dataset, including poverty, food access, cultural factors, and genetic predisposition.
- **RMSE: 2.8374:** The Root Mean Squared Error tells us that, on average, the model's obesity predictions are off by about 2.8 percentage points. Given that obesity rates across counties range from roughly 20% to 50%, an average error of 2.8 points represents good predictive accuracy.

When Hidalgo's actual values for all 8 variables were fed into the model, it predicted an obesity rate of 36.32%. Hidalgo's actual rate is 44.20%, returning a residual of -7.88 percentage points. This means that even a well-fitted national model, fully accounting for Hidalgo's health risk profile, still underestimates its obesity burden by nearly 8 points. This is compelling since it shows that the analyzed variables alone do not tell the whole story when it comes to the obesity percentage of Hidalgo County; other factors not accounted for do so as well.

8. Other Machine Learning Models

Three machine learning classification models were trained to identify high-obesity counties (above the national median of 37.9%) using the 8 health variables. The models were trained on 80% of counties (2,365 counties) and evaluated on the remaining 20% (591 counties).

Model	Accuracy	Precision (High)	Recall (High)	AUC
Logistic Regression	80.0%	0.82	0.76	0.8785
Decision Tree	78.7%	0.83	0.73	0.8616
Random Forest	81.6%	0.83	0.80	0.9097

All three models correctly classify Hidalgo County as high obesity. The Random Forest achieves the highest performance across all metrics — 81.6% accuracy and an AUC of 0.9097, which crosses into the 'excellent' threshold (above 0.90). This means the Random Forest correctly distinguishes high from low obesity counties over 90% of the time using only 8 health variables. The Decision Tree (AUC 0.8616) and Logistic Regression (AUC 0.8785) are both in the 'strong' range. The consistent performance across all three very different model types strengthens confidence that the relationship between these health variables and obesity is real and robust.

ML Models Explanation in Plain Terms:

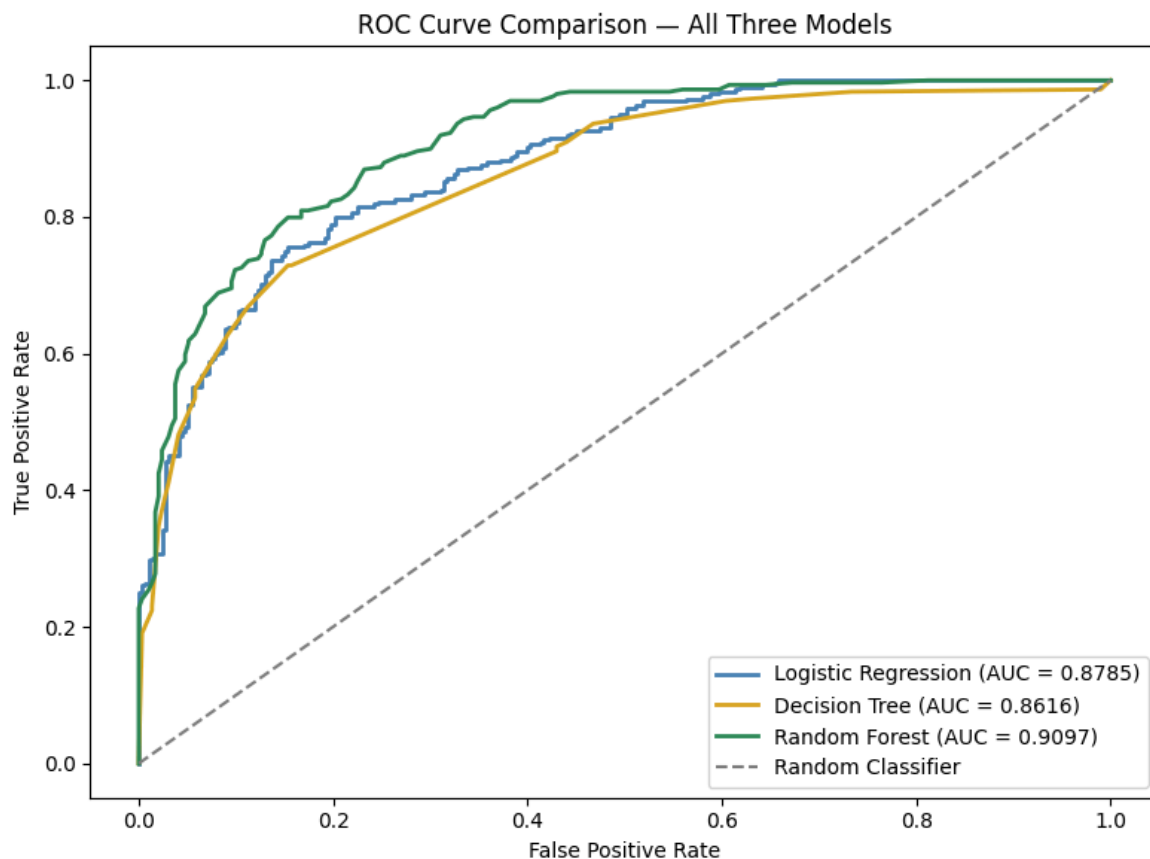
Accuracy is the percentage of counties the model correctly classifies as high or low obesity. An accuracy of 81.6% means that out of every 100 counties, the model correctly identifies 82 of them based solely on their health risk profiles — without ever being told their actual obesity rate.

Precision answers: when the model says a county is high obesity, how often is it right? A precision of 0.83 means 83% of the time. In a clinical analogy, this is like asking how often a positive screening test reflects a true positive.

Recall answers: out of all the counties that actually are high obesity, how many does the model catch? A

recall of 0.80 means it catches 80% of them. Missing the other 20% would be the equivalent of a false negative on a screening test.

AUC (Area Under the ROC Curve) is the most comprehensive single metric. AUC measures the model's ability to rank counties correctly regardless of the classification threshold. Think of it this way: if someone randomly picked one high-obesity county and one low-obesity county and asked the model to identify which is which, the AUC tells you how often it gets it right. An AUC of 0.50 means the model is no better than a coin flip. Above 0.80 is considered strong. Above 0.90 — which our Random Forest achieves at 0.9097 — is considered excellent. The fact that three completely different modeling approaches all achieve strong-to-excellent AUC values on the same data means the findings are not a statistical accident, but rather reflect a genuine, reproducible relationship between these health variables and obesity rates.

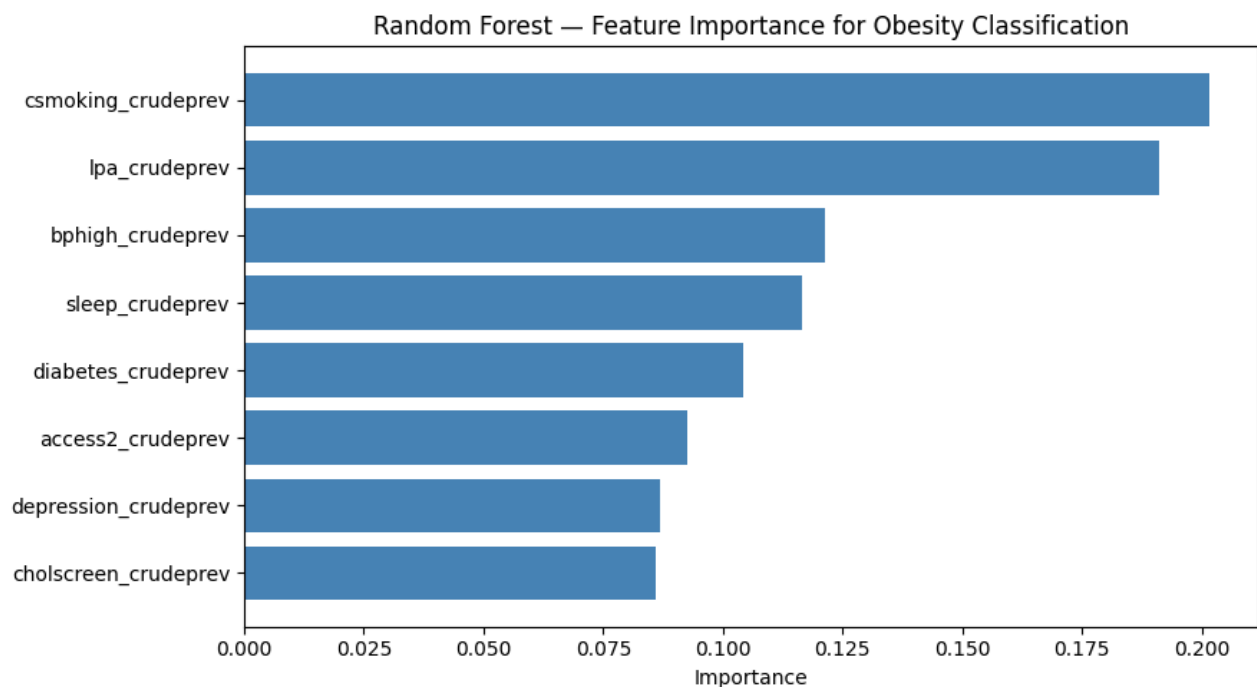


Random Forest Feature Importance

The Random Forest model also identifies which variables were most useful for distinguishing high from low obesity counties:

Rank	Variable	Importance Score	Interpretation
1	Smoking	0.2017	Most predictive variable nationally
2	Physical Inactivity	0.1912	Close second; strongly modifiable

3	High Blood Pressure	0.1212	Downstream cardiovascular marker
4	Short Sleep	0.1164	Underappreciated metabolic risk factor
5	Diabetes	0.1042	Bidirectional with obesity
6	No Insurance	0.0927	Structural barrier to care
7	Depression	0.0869	Behavioral health link
8	Cholesterol Screen	0.0858	Access to care proxy



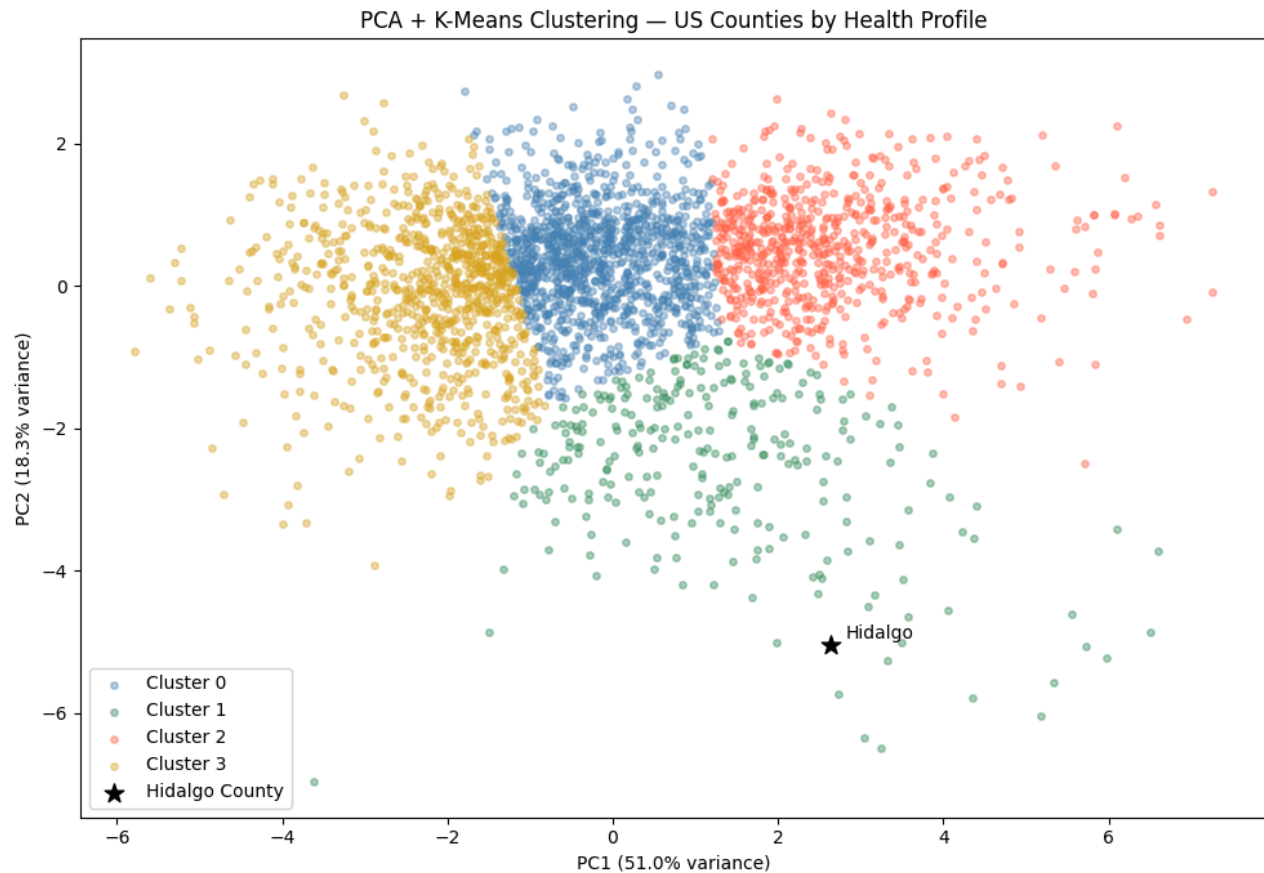
9. PCA & Cluster Analysis

Principal Component Analysis (PCA) was used to compress the 8 health variables into 2 summary dimensions for visualization. The first two principal components captured 69.35% of the total variation in the data (PC1: 51.0%, PC2: 18.3%).

In simple terms, PC1 represents an overall health burden score — counties scoring high on PC1 tend to be unhealthy across multiple dimensions simultaneously. PC2 captures a secondary dimension, distinguishing counties that are unhealthy in different ways (e.g., high smoking vs. high uninsured rates).

K-Means clustering grouped all 3,143 counties into 4 distinct health profiles. Hidalgo County was assigned to Cluster 1, characterized by: the highest uninsured rate of all 4 clusters (20.85% — nearly double the national average), Elevated physical inactivity (32.34%), and above-average obesity (38.57% cluster mean). Critically, Hidalgo's actual obesity rate of 44.2% substantially exceeds even its own cluster's average of

38.57% by a difference of 5.63 percentage points. This means Hidalgo is an outlier even among counties with similar health risk profiles, suggesting that unmeasured factors (poverty, food desert access, or cultural and genetic factors) are contributing additional burden beyond what the 8 variables capture.

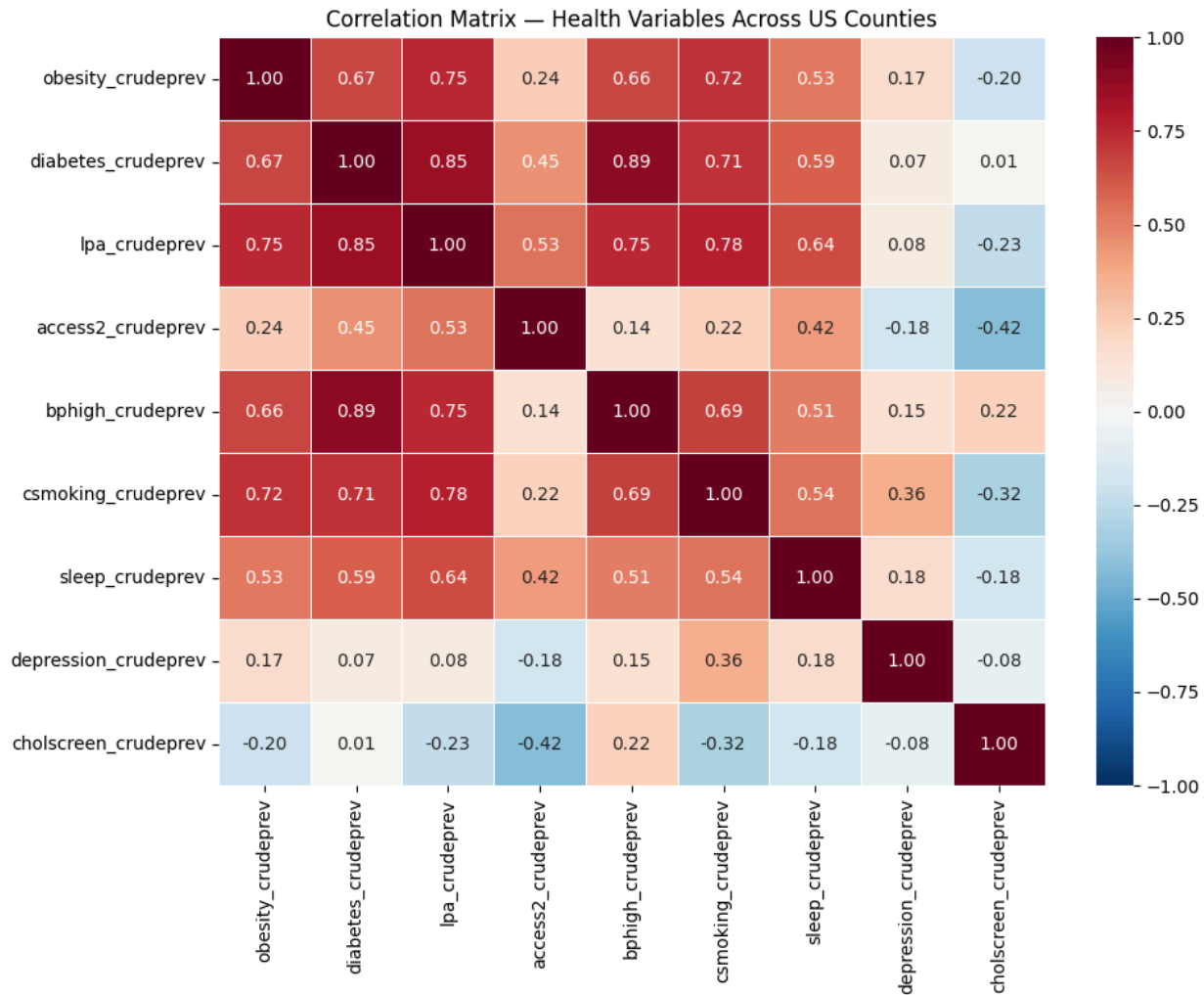


10. Correlation Analysis

Pearson correlation coefficients measure the strength and direction of the linear relationship between each variable and obesity rates across all 3,143 U.S. counties.

Variable	Correlation with Obesity	Interpretation
Physical Inactivity	$r = +0.75$	Strong positive — most correlated single variable
Smoking	$r = +0.72$	Strong positive
Diabetes	$r = +0.67$	Strong positive
High Blood Pressure	$r = +0.66$	Strong positive
Short Sleep	$r = +0.53$	Moderate positive
No Insurance	$r = +0.24$	Weak positive

Depression	$r = +0.17$	Weak positive
Cholesterol Screening	$r = -0.20$	Weak negative



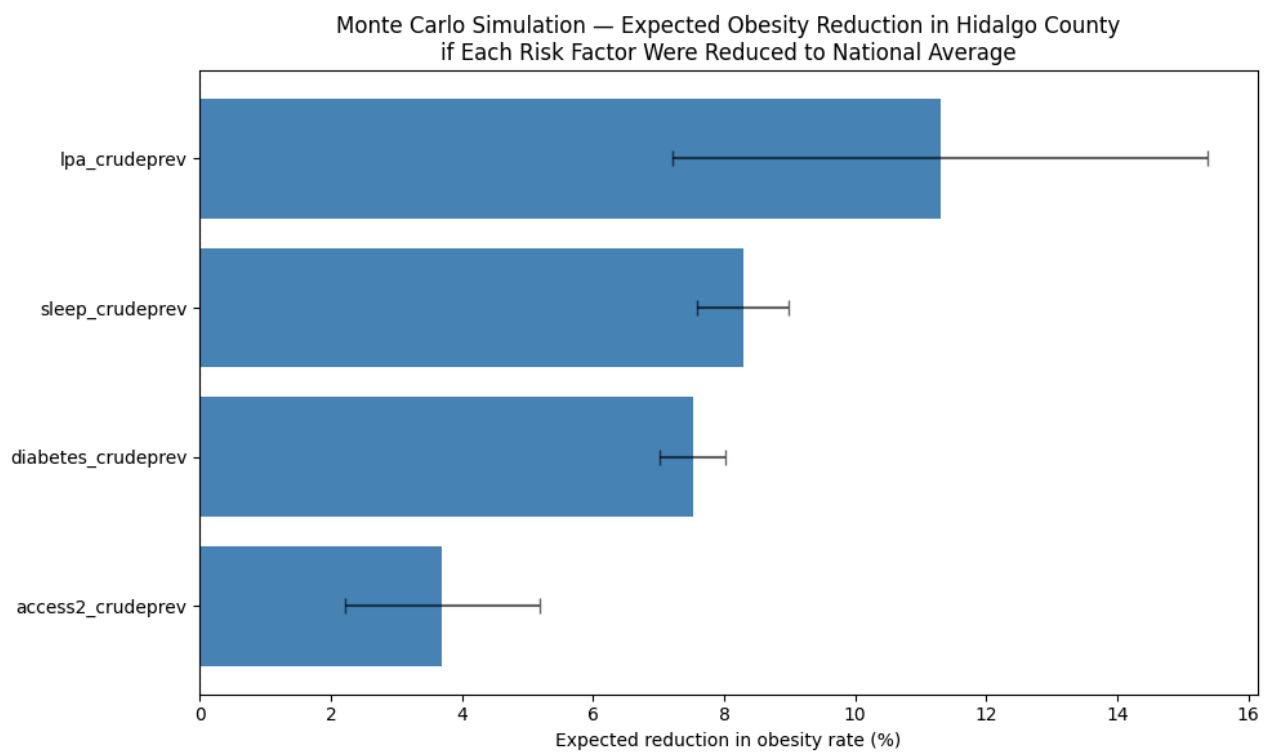
Physical inactivity and smoking are the two strongest correlates of obesity at the county level nationally. The negative correlation of cholesterol screening with obesity is particularly meaningful for clinical practice: counties where more adults receive preventive care screening consistently show lower obesity rates, reinforcing the importance of healthcare access as a protective factor.

11. Monte Carlo Simulation

The linear regression model was used as the basis for a Monte Carlo simulation to estimate the expected reduction in Hidalgo County's obesity rate if specific risk factors were reduced to national average levels. Each simulation ran 10,000 iterations, replacing the target variable with a random draw from the national distribution while holding all other variables at Hidalgo's actual values.

Individual variable simulations

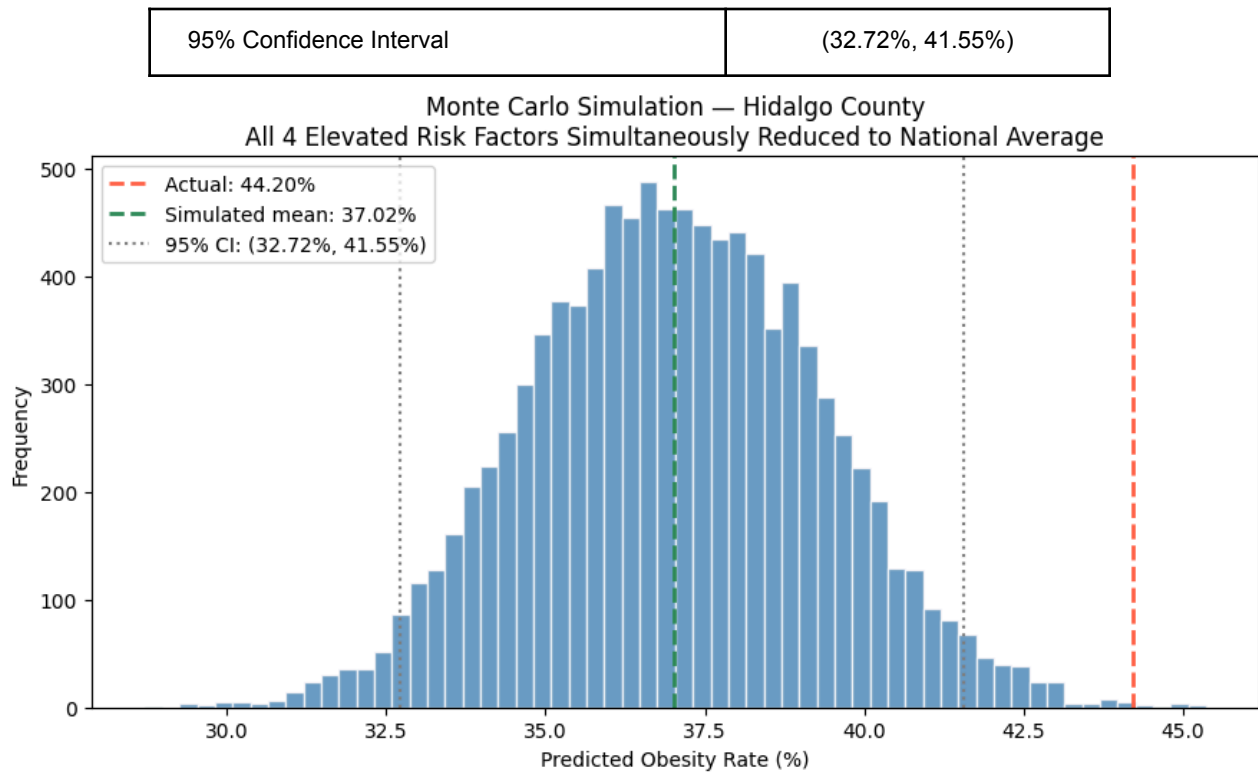
If this variable reduced to national avg...	Predicted obesity	95% Confidence Interval	Expected reduction
Physical inactivity reduced	32.89%	(28.74%, 36.97%)	-11.31%
Short sleep reduced	35.91%	(35.20%, 36.61%)	-8.29%
Diabetes rate reduced	36.67%	(36.16%, 37.18%)	-7.53%
Uninsured rate reduced	40.50%	(39.06%, 42.00%)	-3.70%



Combined simulation — all 4 variables simultaneously

When all four elevated variables are simultaneously reduced to national average levels, the model predicts:

Metric	Value
Hidalgo actual obesity rate	44.20%
Predicted obesity (all 4 reduced)	37.02%
Expected reduction	-7.18%



A reduction of 7.18 percentage points would bring Hidalgo County from the 94th percentile nationally to approximately the national median (37.9%). This would represent a transformative public health outcome for the county.

12. Clinical & Policy Recommendations

Based on the analysis, the following evidence-based recommendations are prioritized by expected impact on obesity reduction in Hidalgo County:

12.1 Priority 1: Physical Activity Programs

Physical inactivity shows the strongest correlation with obesity ($r = 0.75$) and the largest individual Monte Carlo effect (-11.3%). Hidalgo County's inactivity rate of 37.5% is 9 percentage points above the national average.

- Providers should screen all patients for physical activity levels using validated tools and prescribe structured activity as a clinical intervention
- Public health officials should prioritize safe walking infrastructure, parks, and community exercise programs, particularly in colonias and rural areas
- Workplace wellness programs and school-based physical education should be expanded

12.2 Priority 2: Sleep Health Interventions (Expected impact: -8.3%)

Short sleep duration is frequently overlooked as an obesity risk factor, yet Hidalgo County's rate of 40.4% (4.6 points above national) contributes meaningfully to the obesity burden.

- Providers should routinely screen for sleep disorders, particularly obstructive sleep apnea,

- which is highly prevalent in obese patients and creates a vicious cycle
- Patient education should address sleep hygiene as a weight management tool

12.3 Priority 3: Diabetes Management & Prevention (Expected impact: –7.5%)

Hidalgo County's diabetes rate of 17.4% is 3.8% above the national average. The bidirectional relationship between obesity and diabetes means that better diabetes management can reduce obesity, and vice versa.

- Increase prescribing of GLP-1 receptor agonists (e.g., semaglutide) where clinically appropriate, as these now show both glucose control and weight reduction benefits; push for the continued research of other GLP-like peptides
- Continue to push for the expansion and funding of diabetes education programs

12.4 Priority 4: Insurance Coverage Expansion (Expected impact: –3.7%)

Hidalgo County's uninsured rate of 37.4% is more than three times the national average of 11.1%. Uninsured patients cannot access preventive services, medication, or chronic disease management, which creates a structural barrier to every other intervention on this list.

- Aggressive outreach and advocacy efforts for Medicaid enrollment and healthcare plan enrollment

13. Limitations & Caveats

Ecological fallacy: All data in this analysis are at the county level. Essentially, we are comparing counties, not individuals. A correlation between uninsured rates and obesity at the county level does not necessarily mean that uninsured individuals within a county are the ones who are obese.

Missing variables that could have helped paint a better picture: The CDC PLACES dataset does not include poverty rates, food desert access, dietary quality, or genetic predisposition data, at least in the edition reviewed in this study. The linear regression model explains 66% of obesity variation, meaning the remaining 34% is likely attributable to these unmeasured factors. Hidalgo's actual obesity rate (44.2%) being 7.88 percentage points above the model's prediction strongly suggests that poverty and food access might be important unmeasured drivers.

Model-based estimates: CDC PLACES values are model-based estimates, not direct measurements. They are highly reliable for large counties like Hidalgo, but carry more uncertainty for smaller counties.

Monte Carlo assumptions: The simulation assumes the linear regression model accurately captures the causal relationship between risk factors and obesity. In reality, the relationships are complex and non-linear. The simulation should be interpreted as a directional estimate, not a precise forecast.

BMI limitations: Obesity is defined using BMI ≥ 30 , which is a population-level screening tool with known limitations for individual assessment, particularly for Hispanic and indigenous-descent populations where metabolic risk may differ at equivalent BMI values.

14. Conclusion

This analysis provides strong, multi-method evidence that Hidalgo County faces a genuine and severe

obesity crisis. The county ranks in the 94th percentile nationally for obesity prevalence, a difference from the national average that is statistically significant at $p < 0.0001$. Every machine learning model tested correctly identifies Hidalgo as a high-obesity county based solely on its health risk profile.

Four variables stand out as driving Hidalgo's elevated burden above and beyond what is typical even for similar counties: physical inactivity, short sleep duration, diabetes prevalence, and lack of health insurance. Notably, Hidalgo's actual obesity rate is 7.88 percentage points above even what a well-fitted model would predict, suggesting that unmeasured structural factors such as poverty, food desert access, and socioeconomic stress may compound the burden.

The Monte Carlo simulation provides a data-driven roadmap for intervention: if providers and public health officials successfully reduced all four elevated risk factors to national average levels, the model estimates a 7.18 percentage point reduction in obesity prevalence at 95% confidence. This would represent one of the most significant health achievements in the region's history, but it requires coordinated, simultaneous action across clinical practice, community health, and health policy. No single intervention is sufficient.

Providers should screen for physical activity and sleep as well as evaluate insurance status at every visit. These are the primary modifiable drivers of patients' most prevalent and costly chronic condition.

The data argue clearly for a comprehensive, multi-pronged strategy. Physical activity infrastructure, sleep health education, diabetes prevention programs, and insurance enrollment outreach must be pursued simultaneously.

15. GitHub

[Repository](#)